OFFICE 1 | INSTALLATION DETAILS AND DESIGN CONDITIONS

Supplement to normal lighting/power plans and DB schematic. Native editable detail; schematic, not to scale.

1. EVERY POWER ASSEMBLY HAS ONE UPSTREAM 13A FCU

Accessible main/ring junction

13A switched FCU at tap

4mm2 L/N/CPC spur
0.5m assumed

ONE assembly: FB / SO / FS
All outlets share 13A continuous capacity



For rings: both L/N/CPC ends return to their own DB terminals. FCU is before the spur; no unfused extra taps. Main terminals must accept 6/10mm2.

FB01-09: one FCU per four-socket floorbox. SO01-03: one FCU per twin socket. FS01: one switched FCU. 13A is not an instantaneous current clamp.

2. SEPARATE CONTAINMENT BANKS - OVERLAPPING PLAN LINES ARE ROUTING CORRIDORS

Lighting LC-A..G	A: P1/P2	B: P3/P4	C: P5/P6/P8	P7 alone	Feeder alone
7 circuits / Cg0.54	4 ring legs / Cg0.65	4 ring legs / Cg0.65	3 radials / Cg0.70	Cg1.00	Cg1.00

Method B, no insulation contact. Keep L/N/CPC of each circuit together. Between banks provide horizontal cable clearance >2x largest adjacent cable OD

(EMSD A5(3) Note2); separate enclosures and verify cable clearance. A partition alone is not thermal separation. If grouping changes, recalculate sizes.

3. A1 RING DESIGN BASIS AND EQUIPMENT SPECIFICATION

EMSD 6C/6E, Table6(1): B32; 6mm2 copper; corrected leg Iz23.1855A >=20A; each service envelope <=100m2; intact L/N/CPC rings; two fused taps per ring.

Ring leg currents14.658-15.203A support load distribution; overload protection follows the A1 standard arrangement. Verify ring continuity before energisation.

TN-S source Ze0.20ohm/PFC<=6kA are design conditions. Main and RCBO Icn10kA. Required total I2t: B10/B16<=70000; B32/FCU<=90000; C63<=1000000 A2s.

Select apparatus meeting the stated breaking capacity and total clearing energy. Control20AX/LED>=600W; confirm driver inrush. PE bypasses switching.

4. FINAL LIGHTING BASIS - COURSE GEOMETRY AND HEIGHT ASSUMPTIONS

92 panels /2256W. LDT calculation, MF0.80, luminaires1.9m above workplane: mean direct lux open592.9 /pantry546.3 /upper east588.5 /south east662.6.

South east: four38W panels,4685.55lm each; mean589.2lx at2.2m above workplane,17.8% above500lx target. Lighting positions and routes retained.

Adopted height1.9m and MF0.80 with periodic cleaning. Direct-light calculation; see report for room geometry and perimeter illuminance.